

- 3 (a) straight line through origin
 negative gradient [1]
- (b) $a = -\omega^2 x$ and $\omega = 2\pi f$ C1
 $750 = (2\pi f)^2 \times 0.3 \times 10^{-3}$ C1
 $f = 250 \text{ Hz}$ A1 [3]
- (c) straight line between $(-0.3, +190)$ and $(+0.3, -190)$ A2 [2]
(allow 1 mark for end of line incorrect by one grid square or line does not extend to $\pm 0.3 \text{ mm}$)

[Total: 7]

- 4 (a) charge / potential (ratio must be clear) B1 [1]